Dijkstra's Algorithm:

Finding the Shortest Path

introduction to a
foundational graph algorithm



What is Dijkstra's Algorithm?

Finds shortest paths from a source node to all other nodes in a weighted graph

Works with graphs that have non-negative edge weights

Widely used in routing, navigation, and network optimization



How Dijkstra's Algorithm Works

01

Initialize

Set the source distance to 0 and all other nodes to infinity

02

Select Closest

Use a priority queue to pick the unvisited node with the smallest distance

03

Relax Edges

Update neighbor distances if a shorter path is found

04

Repeat

Repeat until all nodes are visited or the shortest paths are finalized



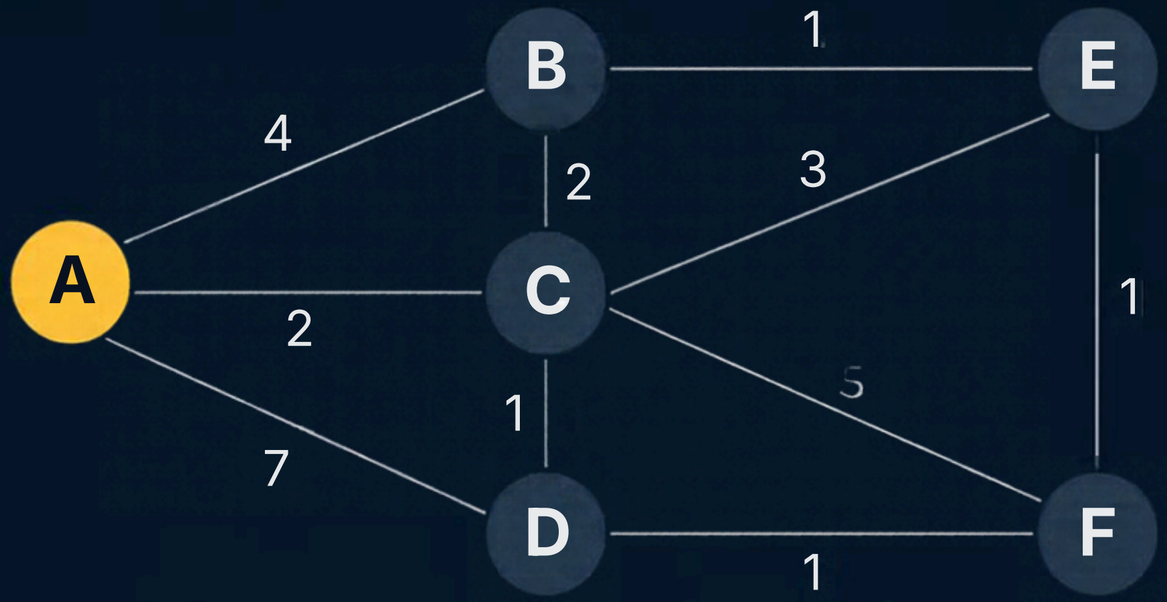
Example Walkthrough

Start at source **A**

Pick closest unvisited node from the priority queue

Relax edges and **update** shortest distances

Repeat until all nodes are **visited**



Step	Picked Node	Priority Queue (next)	Distances from A
1	A	C (2), B (4), D (7)	A: 0 B: 4 C: 2 D: 7 E: ∞ F: ∞
2	C	B (4), D (3), E (5), F (7)	A: 0 B: 4 C: 2 **D: 3** E: 5 F: 7
3	D	B (4), E (5), F (4)	A: 0 B: 4 C: 2 D: 3 E: 5 **F: 4**
4	B	E (5), F (4)	A: 0 B: 4 C: 2 D: 3 E: 5 F: 4
5	F	E (5)	A: 0 B: 4 C: 2 D: 3 E: 5 F: 4
6	E	–	A: 0 B: 4 C: 2 D: 3 **E: 5** F: 4

Applications of Dijkstra's Algorithm

GPS navigation and route planning

Network routing protocols

Urban traffic management

Robotics pathfinding



Summary and Questions

Efficiently finds shortest paths in weighted graphs

Powers routing, navigation, and network optimization

A foundational algorithm in computer science

